

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Year: 2012

Subject: 311131 Mechanical Drawing

Section: 5-6

Date: 2 January 2013

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. **Dictionary and calculator** are NOT allowed in the examination.
4. This is a closed book exam (textbooks and lecture notes are NOT allowed during the exam).
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 8 pages (including this page), 8 problems and a total score of 60 marks.
7. Write all your answers in this examination sheet.

Problem 1 Answer all these questions**(10 marks)**

➤ Lettering is usually UPPERCASE (True or False) (1 mark)

➤ List the six principal views of orthographic projection. (2 marks)

1.

2.

3.

4.

5.

6.

➤ In a drawing that shows the top, front, and right-side view which two views show depth?
Which view shows depth vertically on the drawing sheet? Which view shows depth horizontally
on the drawing sheet? (2 marks)

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.....

➤ Most engineering lettering is (select the correct answer) (1 mark)

a) Roman

b) Gothic

c) Italic

d) Old English

e) Script

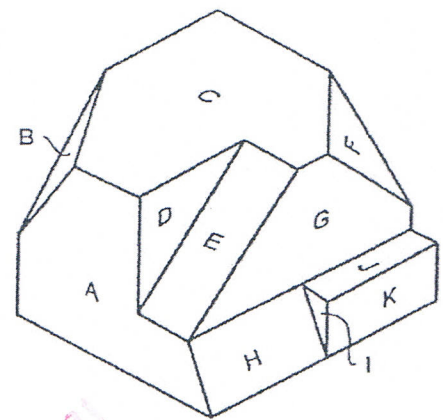
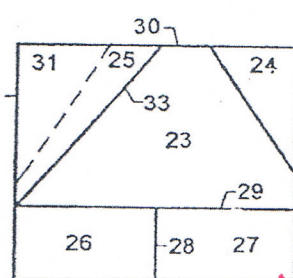
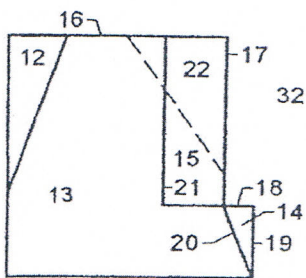
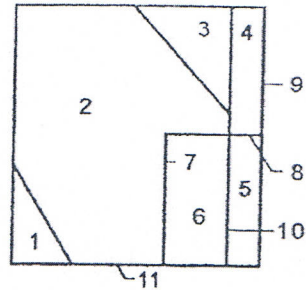
➤ What can the draft-person do to prevent confusion between first and third angle projection? (2 marks)

.....

.....

➤ A straight line always projects as a or as a (2 marks)

Problem 2 From given 3rd angle orthogonal projection views and pictorial (isometric) drawing, match the surface numbers and surface A-K, as an example of surface A. (10 marks)



Surface	Front view	Side view	Top view
A	13	32	11
B			
C			
D			
E			
F			
G			
H			
I			
J			
K			

Problem 3 The same objects were placed in different view, I and II. Match the lines between both views. (10 marks)

	A		
	B		
	C		
	D		
	E		
	F		

	A		
	B		
	C		
	D		
	E		
	F		

	A		
	B		
	C		
	D		
	E		
	F		

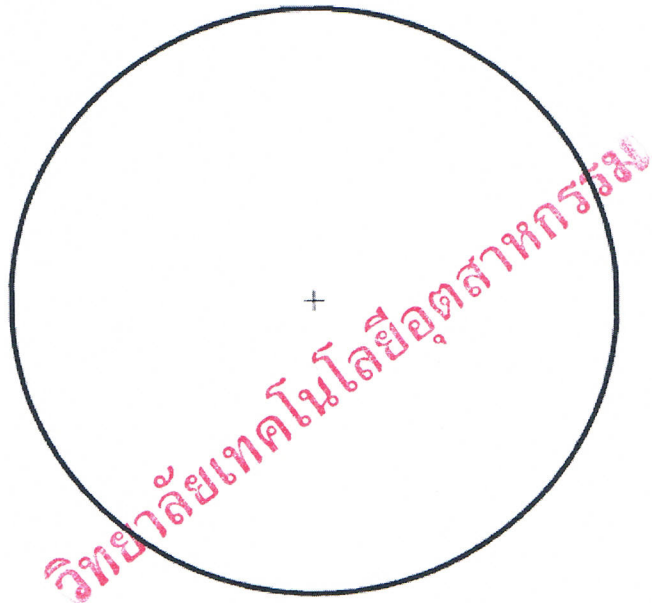
	A		
	B		
	C		
	D		
	E		
	F		

Instructions: (for Problem 4)

- The following exams must be constructed with the aid of compasses and set squares.
- Ensure to use correct and accurate methods.
- All construction lines and projection lines must be shown.
- A uniform thickness and darkness of visible line is required throughout.
- Emphasis is placed on creating drawings that are neat, correctly shape and scale.

Problem 4 Construct a hexagon inside the circle.

(5 marks)



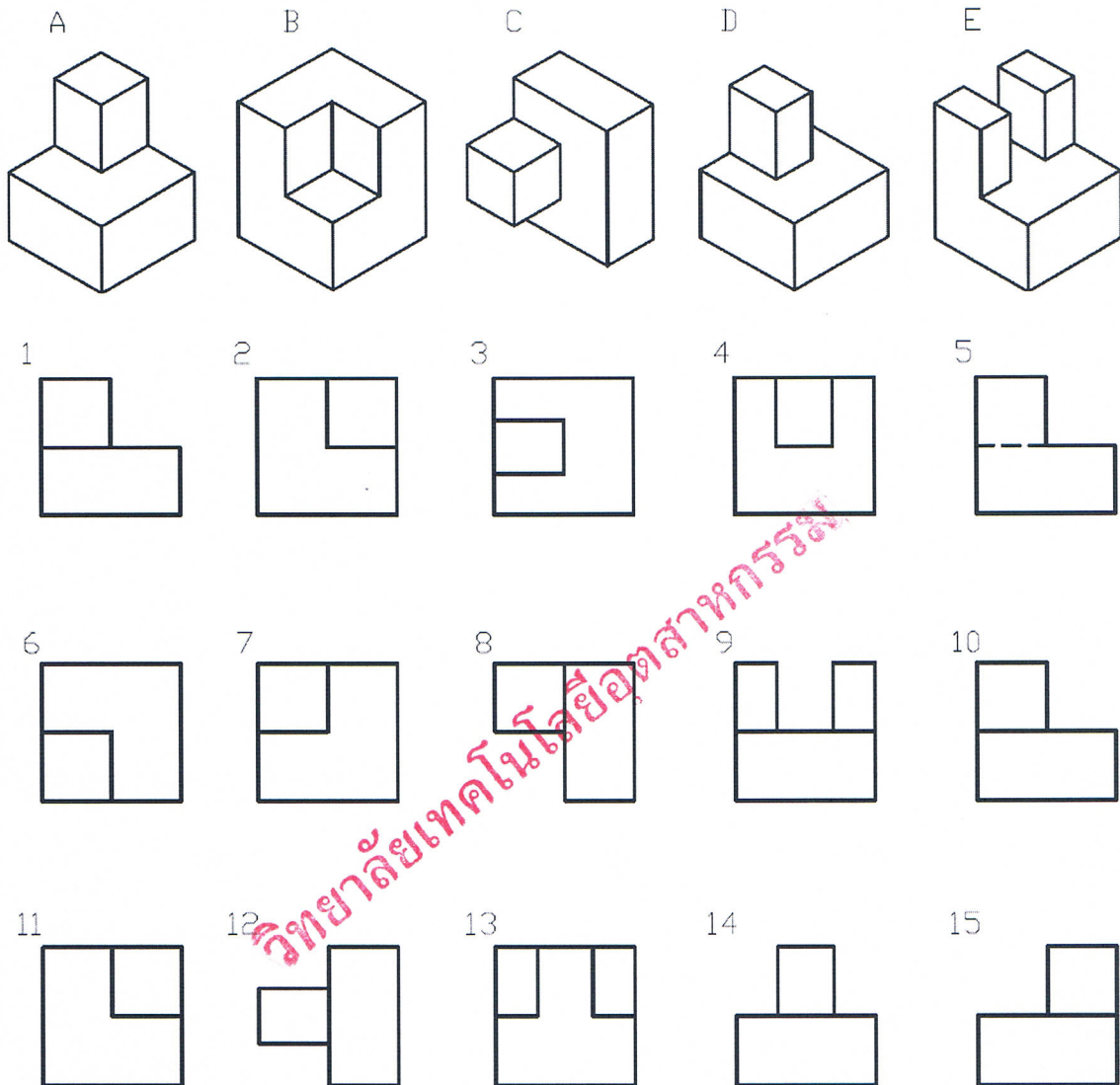
Construct an arc with a radius of 30 mm tangential to given lines.

(5 marks)



Problem 5 From given isometric drawing, select front, top and side views.

(5 marks)

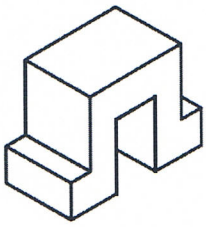
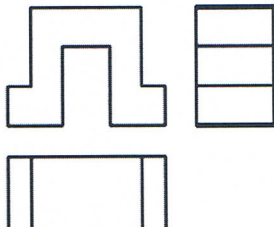
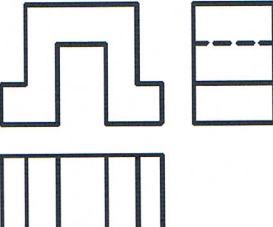
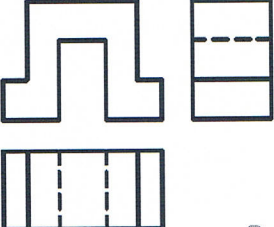
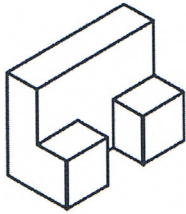
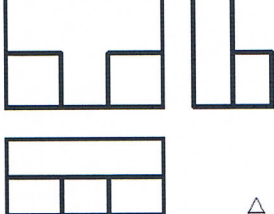
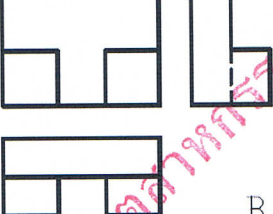
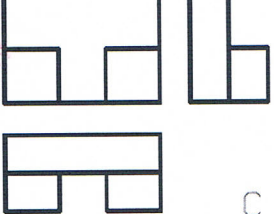
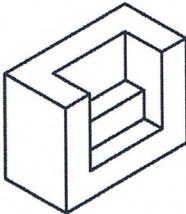
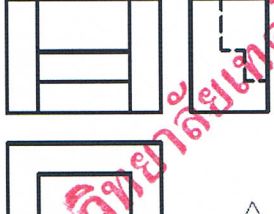
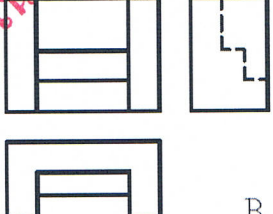
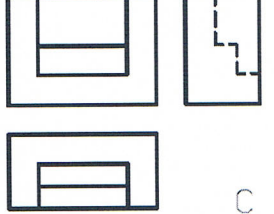
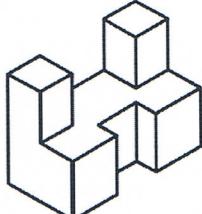
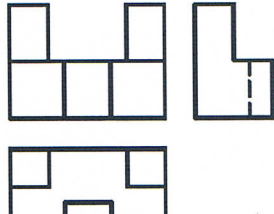
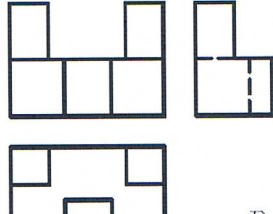
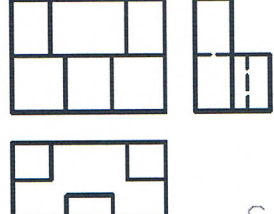
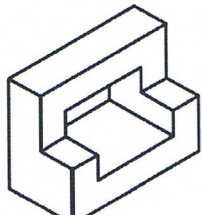
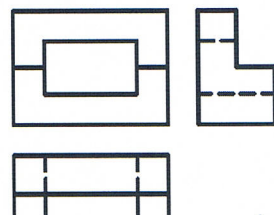
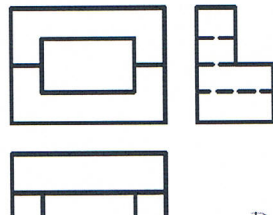
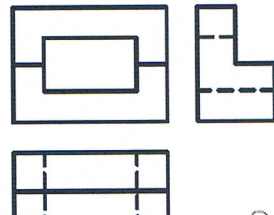


ISOMETRIC DRAWING	A	B	C	D	E
FRONT VIEW					
TOP VIEW					
SIDE VIEW					

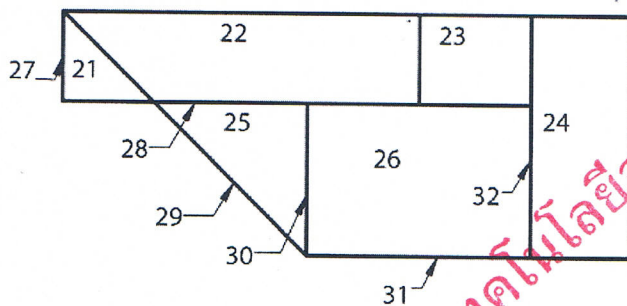
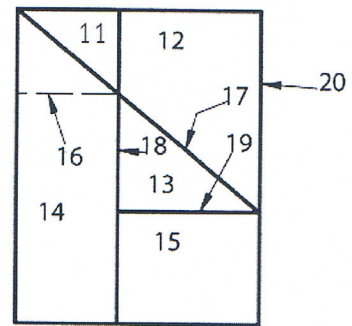


Problem 6 Select the multiview drawing from given isometric drawing.

(5 marks)

1				
2				
3				
4				
5				



[illegible]